

Francesco Emanuele Stradi

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ABOUT ME

I am a postdoctoral researcher at Politecnico di Milano, working with Prof. Alberto Marchesi. I received my Ph.D. in computer science at Politecnico di Milano, where I was advised by Prof. Nicola Gatti. My current research focuses on the theoretical aspects of artificial intelligence, particularly online reinforcement learning (RL) and online learning. Specifically, I am interested in combining online (machine) learning theoretical tools with optimization techniques to build strategic agents able to act in complex online environments.

EXPERIENCE

- **Politecnico di Milano** May 2026 - now
Postdoctoral researcher
◦ Working with Prof. Alberto Marchesi
Milan, Italy
- **lastminute.com** June - September 2024
Machine Learning Scientist, internship
◦ In charge of developing online learning algorithms for dynamic pricing
Chiasso, Switzerland
- **Polimi Artificial Intelligence Research and Innovation Center (Airic)** August - October 2022
Research Scientist, internship
◦ In charge of selecting State-Of-The-Art artificial intelligence techniques and applying them in real world business oriented scenarios
Milan, Italy

EDUCATION

- **Politecnico di Milano** 2022 - 2026
PhD in Computer Science
◦ Advised by Prof. Nicola Gatti
◦ Conferred *cum laude*
Milan, Italy
- **Columbia University** February - April 2025
Visiting Scholar
◦ Working with Prof. Christian Kroer at IEOR Department
New York City, USA
- **Politecnico di Milano** 2019 - 2022
Msc in Computer Science and Engineering
◦ Grade: 110 cum laude/110
Milan, Italy
- **Politecnico di Milano** 2016 - 2019
Bsc in Computer Engineering
Milan, Italy

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, W=WORKSHOP, T=THESIS

- [C.1] Kalana Kalupahana*, Francesco Emanuele Stradi*, Matteo Castiglioni and Alberto Marchesi (2026). **Toward Optimal Regret in Robust Pricing: Decoupling Corruption and Time**. In *The 40th Conference on Neural Information Processing Systems [NeurIPS 2026]*.
- [C.2] Francesco Bacchiocchi*, Matteo Castiglioni*, Alberto Marchesi* and Francesco Emanuele Stradi* (2026). **Multi-Armed Bandits With Best-Action Queries**. In *The 40th Conference on Neural Information Processing Systems [NeurIPS 2026]*.
- [C.3] Eleonora Fidelia Chiefari, Francesco Emanuele Stradi, Matteo Castiglioni and Alberto Marchesi (2026). **Online Resource Allocation With General Constraints**. In *The 40th Conference on Neural Information Processing Systems [NeurIPS 2026]*.
- [C.4] Matteo Bollini*, Gianmarco Genalti*, Francesco Emanuele Stradi*, Matteo Castiglioni and Alberto Marchesi (2026). **Replicable Constrained Bandits**. In *The 40th Conference on Neural Information Processing Systems [NeurIPS 2026]*.
- [C.5] Francesco Emanuele Stradi, Kalana Kalupahana, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2026). **Truly Adapting to Adversarial Constraints in Constrained MABs**. In *The 39th Annual Conference on Learning Theory [COLT 2026]*.
- [C.6] Francesco Emanuele Stradi, Matteo Castiglioni, Alberto Marchesi, Nicola Gatti and Christian Kroer (2025). **No-Regret Learning Under Adversarial Resource Constraints: A Spending Plan Is All You Need!**. In *The 39th Conference on Neural Information Processing Systems [NeurIPS 2025]*.

- [C.7] Gianmarco Genalti*, **Francesco Emanuele Stradi***, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Data-Dependent Regret Bounds for Constrained MABs**. In *The 39th Conference on Neural Information Processing Systems [NeurIPS 2025]*.
- [C.8] **Francesco Emanuele Stradi**, Anna Lunghi, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Taming Adversarial Constraints in CMDPs**. In *The 39th Conference on Neural Information Processing Systems [NeurIPS 2025]*.
- [C.9] Francesco Bacchiocchi*, **Francesco Emanuele Stradi***, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Markov Persuasion Processes: Learning to Persuade From Scratch**. In *The 39th Conference on Neural Information Processing Systems [NeurIPS 2025]*.
- [C.10] **Francesco Emanuele Stradi**, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Learning Adversarial MDPs with Stochastic Hard Constraints**. In *The 42nd International Conference on Machine Learning [ICML 2025]*.
- [C.11] **Francesco Emanuele Stradi**, Anna Lunghi, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Policy Optimization for CMDPs with Bandit Feedback: Learning Stochastic and Adversarial Constraints**. In *The 42nd International Conference on Machine Learning [ICML 2025]*.
- [C.12] **Francesco Emanuele Stradi**, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Optimal Strong Regret and Violation in Constrained MDPs via Policy Optimization**. In *The 13th International Conference on Learning Representations [ICLR 2025]*.
- [C.13] Davide Maran*, Francesco Bacchiocchi*, **Francesco Emanuele Stradi***, Matteo Castiglioni, Nicola Gatti and Marcello Restelli (2024). **Bandits with Ranking Feedback**. In *The 38th Conference on Neural Information Processing Systems [NeurIPS 2024]*.
- [C.14] **Francesco Emanuele Stradi**, Jacopo Germano, Gianmarco Genalti, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2024). **Online Learning in CMDPs: Handling Stochastic and Adversarial Constraints**. In *The 41st International Conference on Machine Learning [ICML 2024]*.
- [C.15] Francesco Bacchiocchi*, **Francesco Emanuele Stradi***, Matteo Papini, Alberto Maria Metelli and Nicola Gatti (2024). **Online Learning with Off-Policy Feedback in Adversarial MDPs**. In *The 33rd International Joint Conference on Artificial Intelligence [IJCAI 2024]*.
- [C.16] Davide Maran*, Pierricardo Olivieri*, **Francesco Emanuele Stradi***, Nicola Gatti and Marcello Restelli (2024). **Online Markov Decision Processes Configuration with Continuous Decision Space**. In *The 38th AAAI Conference on Artificial Intelligence [AAAI 2024]*.
- [J.1] **Francesco Emanuele Stradi**, Anna Lunghi, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2026). **Policy Optimization for CMDPs with Bandit Feedback: Best-of-Both-Worlds and Beyond**. In *Artificial Intelligence [AIJ 2026]*.
- [W.1] Gianmarco Genalti*, **Francesco Emanuele Stradi***, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Data-Dependent Regret Bounds for MABs with Constraints**. In *NeurIPS 2025 Workshop: Aligning Reinforcement Learning Experimentalists and Theorists [ARLET 2025]*.
- [W.2] **Francesco Emanuele Stradi**, Anna Lunghi, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Policy Optimization in CMDPs with Bandit Feedback: Learning with Stochastic and Adversarial Constraints**. In *NeurIPS 2025 Workshop: Aligning Reinforcement Learning Experimentalists and Theorists [ARLET 2025]*.
- [W.3] **Francesco Emanuele Stradi**, Anna Lunghi, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Taming Adversarial Constraints in Constrained MDPs**. In *The 18th European Workshop on Reinforcement Learning [EWRL 2025]*.
- [W.4] **Francesco Emanuele Stradi**, Anna Lunghi, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Policy Optimization for CMDPs with Bandit Feedback: Learning Stochastic and Adversarial Constraints**. In *The 18th European Workshop on Reinforcement Learning [EWRL 2025]*.
- [W.5] **Francesco Emanuele Stradi**, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2024). **Online learning in CMDPs with adversarial losses and stochastic hard constraints**. In *The 17th European Workshop on Reinforcement Learning [EWRL 2024]*.
- [W.6] Francesco Bacchiocchi*, **Francesco Emanuele Stradi***, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2024). **Markov Persuasion Processes: How to Persuade Multiple Agents From Scratch**. In *ICML 2024 Workshop: Aligning Reinforcement Learning Experimentalists and Theorists [ARLET 2024]*.
- [W.7] Francesco Bacchiocchi*, **Francesco Emanuele Stradi***, Matteo Papini, Alberto Maria Metelli and Nicola Gatti (2023). **Online Adversarial MDPs with Off-Policy Feedback and Known Transitions**. In *The 16th European Workshop on Reinforcement Learning [EWRL 2023]*.
- [W.8] Davide Maran*, Pierricardo Olivieri*, **Francesco Emanuele Stradi***, Nicola Gatti and Marcello Restelli. (2023). **Online Configuration in Continuous Decision Space**. In *The 16th European Workshop on Reinforcement Learning [EWRL 2023]*.

[T] **Francesco Emanuele Stradi** (2026). **Regret Minimization in Constrained Markov Decision Processes**.
External Reviewers: Prof. Haipeng Luo, Matteo Pirodda, PhD.
* stands for equal contribution

TEACHING

- **Probabilità e Statistica Matematica, TA** *Spring 2024*
Politecnico di Milano
 - Bsc in Management Engineering
- **Statistica, TA** *Spring 2024*
Politecnico di Milano
 - Bsc in Mechanical Engineering

SERVICE

- **Reviewing:** ICLR (2025, 2026), ICML (2025, 2026 **Silver Reviewer**), NeurIPS (2024, 2025, 2026), Mathematics of Operations Research (2025), Transactions on Machine Learning Research (2026), ITCS (2027)
- **Program Committee:** EC (2026)

PROGRAMMING

- **Programming Languages:** C, Java, Python, SQL, Ampl

ADDITIONAL INFORMATION

- **Languages:** Italian (Native), English (Proficiency level)

REFERENCES

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